

CS570 Spring 2025: Analysis of Algorithms Exam III

	Points		Points
Problem 1	20	Problem 4	20
Problem 2	9	Problem 5	20
Problem 3	16	Problem 6	15
Total 100			

First name	
Last Name	
Student ID	

Instructions:

1. This is a 2-hr exam. Closed book and notes. No electronic devices or internet access.
2. A single double sided 8.5in x 11in cheat sheet is allowed.
3. If a description to an algorithm or a proof is required, please limit your description or proof to within 150 words, preferably not exceeding the space allotted for that question.
4. No space other than the pages in the exam booklet will be scanned for grading.
5. If you require an additional page for a question, you can use the extra page provided within this booklet. However please indicate clearly that you are continuing the solution on the additional page.
6. Do not detach any sheets from the booklet.
7. If using a pencil to write the answers, make sure you apply enough pressure, so your answers are readable in the scanned copy of your exam.
8. Do not write your answers in cursive scripts.
9. This exam is printed double sided. Check and use the back of each page.

1) 20 pts (2 pts each)

Mark the following statements as **TRUE** or **FALSE** by circling the correct answer. No need to provide any justification.

[**TRUE/FALSE**]

In any graph, the size of a minimum Vertex Cover plus the size of a maximum Independent Set equals the number of vertices.

[**TRUE/FALSE**]

if $A \leq_p B$ and $A \in \text{NP}$, then $B \in \text{NP}$.

[**TRUE/FALSE**]

There exist problems X where $X \in \text{NP-hard}$ and $X \notin \text{NP}$

[**TRUE/FALSE**]

If a problem is NP-complete, it must have a polynomial-time certifier.

[**TRUE/FALSE**]

NP-Intermediate problems are those problems that neither have polynomial time solutions nor are they polynomial time reducible to NP-complete problems.

[**TRUE/FALSE**]

If we find a $\frac{1}{2}$ - approximation to the independent set problem, we have proven that $P = \text{NP}$.

[**TRUE/FALSE**]

Assume $P \neq \text{NP}$. Suppose problem X can be solved by an algorithm that first formulates an Integer Linear Program (ILP) in polynomial time and then solves this ILP formulation to find the solution to X . This means that X must be NP-hard.

[**TRUE/FALSE**]

Every LP has an optimal solution.

[**TRUE/FALSE**]

A ρ -approximation for the TSP with triangle inequalities, with $\rho < 1.5$ would imply that $P = \text{NP}$.

[**TRUE/FALSE**]

If X is NP-hard and $X \leq_p Y$, then Y cannot belong to NP.

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2) 9 pts - 3 pts each. Select all correct answers. No partial credit I. For

decision problems A and B, suppose $A \leq B$. Assuming $P \neq NP$,

- A) If B is in P, then A is in P.
- B) If A is NP-hard, then B is NP-hard.
- C) If B is NP-hard and belongs to NP, then A is NP-complete.
- D) If B is NP-complete, then A is NP-hard.

II. In an undirected graph with n vertices, what is the maximum possible number of edges that two distinct Hamiltonian Cycles can share?

- A) $\log_2 n$
- B) $n/2$
- C) $n-2$
- D) $n-1$

III. Which of the following statements is true about Linear Programming

- A) There exists a weakly polynomial time algorithm to solve Linear Programming.
- B) If we can solve 3-SAT using LP in weakly polynomial time, we have proven that $P=NP$.
- C) The Max flow problem with integer capacities can be solved using LP.
- D) The Max flow problem with non-integer capacities can be solved using LP.

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3) 16 pts

Suppose USC wants to send teams of students into a coding competition (again, yay!). Eligible students are a set of n freshmen $\{s_1 \dots s_n\}$ and k sophomores $\{s_{n+1} \dots s_{n+k}\}$ who need to be matched to form teams. Each team must have exactly one freshman and one sophomore and no student can be on multiple teams.

Each pair of freshman s_i ($i = 1, \dots, n$), and sophomore s_j ($j = n+1, \dots, n+k$) have a known conflict quotient of C_{ij} . To avoid negativity and chaos, we want to keep the combined conflict quotient of all the teams to at most C^* .

Formulate an Integer Linear Program (ILP) that will compute the maximum number of teams USC can send as per the constraints above.

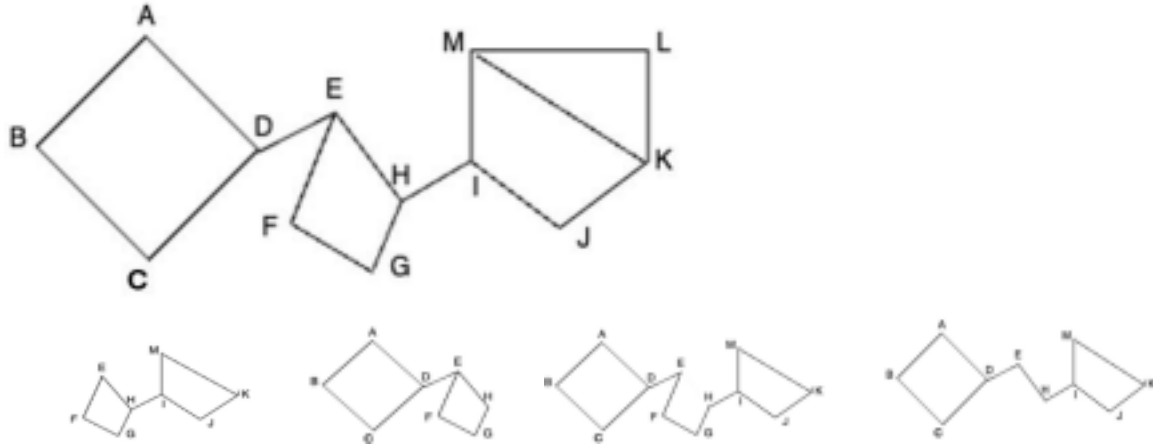
a) Define the decision variables of the ILP (do NOT describe the known input parameters). (4 points)

b) What is the objective of the ILP? (4 points)

c) What are the constraints of the ILP? (8 points)

4) 20 pts

The Dumbbell Containment Problem (DCP) asks: Given an undirected graph $G = (V, E)$, and an integer $k > 2$, is there a k -dumbbell in G . A k -dumbbell consists of two disjoint simple cycles of size k connected by a simple path. A sample graph with its 4 distinct 4-dumbbells are shown below.



Show that DCP is
a) in NP. (5 points)

b) NP-hard. (15 points)

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5) 20 pts

A Doctor and 99 Assistants have opened a laboratory startup to produce special medicine and bandages which require the collaboration of robots and humans. To produce an ounce of medicine, it takes 0.2 hours of human labor and 4 hours of robot labor. To produce an inch of bandage, it takes 0.5 hours of human labor and 2 hours of robot labor. An ounce of medicine sells for \$30 and an inch of bandage sells for \$20. Medicine and bandages can be sold in fractions of an ounce or inch respectively.

They want to maximize profit for the coming month so they can hire more staff. However, the Assistants are busy with other work so they can only devote 20 hours each to this project during the whole month. In addition, the doctor can only devote a total of 3800 robot hours for the month to this project since they are needed for other tasks. Formulate a linear program to help maximize the laboratory profit for the month, and to compute how much of each product to produce given the available human and robot hours. (No need to compute numerical answers to the LP)

A) Describe your LP variables (6 pts)

B) Present your objective function (4 pts)

C) Constraints: (10 pts)

6) 15 points

Recall the Maximum Independent Set (MIS) problem: For an undirected graph $G = (V, E)$, S is an independent set of vertices if no two vertices in S share an edge in E . The MIS problem asks for the independent set with maximum size. We say S is a **Saturated** Independent Set (SIS) if no vertex v can be added to S so that $S \cup \{v\}$ is also an independent set.

For example, suppose graph G is simply a path $ABCDE$. Then, $\{B, D\}$, $\{B, E\}$ and $\{A, C, E\}$ are some examples of SIS, while $\{A, C\}$, $\{C, E\}$, and $\{B\}$ are examples of independent sets that are not saturated.

a) (5 points) Describe an efficient algorithm that outputs an SIS for a given graph G .
(Hint: *something very simple* should suffice)

Now, consider any such algorithm A that computes an arbitrary SIS (i.e., no additional property to its output other than the fact that it is an SIS). We want to analyze if this can serve as a good approximation to the MIS problem.

b) (2 points) Does there exist a constant $0 < \alpha < 1$, s.t., A is guaranteed to output an independent size of $\alpha * M(G)$ on every graph G , where $M(G)$ denotes the size of a MIS in G . (Simply answer Yes or No)

c) (8 points) If you answered Yes in part b), provide a value of α and prove the approximation guarantee. If you answered No, prove that no such α exists.